

Fig. 4: Screenshot of program output showing amplification of photons

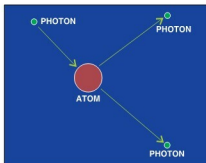


Fig. 5: Program output showing release of photons

One way a laser can measure distance is by time of flight. The other method makes use of an interferometer.

Military. In military, IR counter-measure systems use lasers to confuse the seeker heads on heat-seeking anti-aircraft missiles. High-power, boost-phase intercept laser systems use a complex system of lasers to find, track and destroy intercontinental ballistic missiles.

Some weapons simply use a laser to disorient a person. One such weapon is the Thales green laser optical warner.

Another military use of lasers is as a laser target designator. This is a low-power laser pointer used to indicate a target for a precision-guided munition, typically launched from an aircraft.

Cutting tools. Lasers produce such an intense and precisely focused energy that they can cut through metals, ceramics, plastics and cloths. They have become popular in many industrial operations because high-precision, computer-controlled lasers are much more accurate than human-operated cutting tools and, unlike

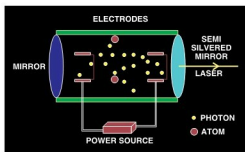


Fig. 6: Program output showing the internal working of a laser

traditional tools, laser beams never turn blunt.

Scientific research. Since the laser was patented in 1958, it has become smaller, more precise and more powerful today. Nuclear fusion, spectroscopy and cooling atoms are some of the laser applications in scientific research.

Microscopy. Confocal laser scanning microscopy and two-photon excitation microscopy make use of lasers to obtain blur-free images of thick specimen at various depths. Laser-capture micro-dissection uses lasers to procure specific cell populations from a tissue section under microscopic visualisation.

Software program

The above laser concepts have been taken into consideration while developing this program. The program is written in 'C' language and compiled using Turbo C integrated development environment. Turbo C has a good collection of graphics libraries. This program uses various 'C' graphics files including egavga.bgi, scri.chr, trip.chr and tscr.chr. All these files should be stored in the same folder as the executable file.

The different program outputs depict working of a laser as below:

Amplification of photons. The first program output screen demonstrates generation of multiple photons inside the laser due to amplification from a single photon (see Fig. 4).

Release of photons. The second screen demonstrates the stimulation process of an atom where a photon stimulates the atom to release more photons (refer Fig. 5). The arrows

have been added in the actual screenshot to show the flow taking place.

Inside the laser. The third screen (Fig. 6) demonstrates the complete process that takes place inside the laser. Proper flow is presented in which photons energise by the application of power supply, amplify with stimulation of the atoms and exit the laser through the semi-silvered mirror. The symbols of the photon and the atom are displayed on the top left.

End of demo. The final screen prompts the user to press any key to exit the program.

Main functions used

Initgraph(&gd,&gm,"c:\\turbo3\\bgi"). Initialises the graphics system by loading a graphics driver from the disk and then putting the system into graphics mode. It also resets all the graphics settings (like colour, palette and current position) to their default values.

***gd.** Specifies the graphics driver to be used.

***gm.** Specifies the initial graphics mode, unless *gd = DETECT. When *gd = DETECT, initgraph sets the graphics mode to the highest resolution available for the detected driver.

Closegraph(). Switches the screen back from graphics mode to text mode. It is called after getch() function so that the screen doesn't clear until the user hits the key.

Void projectwork1(). Demonstrates amplification of photons.

Void projectwork2(). Demonstrates stimulation of an atom to release photons.

Void projectwork3(). Demonstrates how laser works.

Void closing. Provides an exit screen to exit the program.

EFY note. The source code and relevant files of this project are available on www.efymag.com website. ●

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